

Tool Commands

MoveWithSpeed

Controls the movement of the robot's omnidirectional chassis using velocity commands.

```
robot_control call-tool MovewithSpeed --linear-x <value> --linear-y <value> --angular-z <value> --duration <value>
```

Parameter	Type	Required	Description
--linear-x	number	no	Linear velocity in the X direction
--linear-y	number	no	Linear velocity in the Y direction
--angular-z	number	no	Angular velocity around the Z-axis
--duration	number	no	Duration of movement (in seconds)

Move

Controls the robot's chassis to move a specified distance.

```
robot_control call-tool Move --distance <value> --direction <value>
```

Parameter	Type	Required	Description
--distance	number	yes	Movement distance (in meters)
--direction	string	no	Movement direction: 'forward', 'backward', 'left' (strafe), 'right' (strafe)

GetCurrentArmJointAngles

Retrieves the current joint angles of the robotic arm.

```
robot_control call-tool GetCurrentArmJointAngles
```

GetEndEffectorPose

Retrieves the pose (position and orientation) of the end effector (including the gripper and camera).

```
robot_control call-tool GetEndEffectorPose
```

ControlSixArmJoint

Simultaneously controls all 6 joints of the robotic arm; Joint 6 controls the gripper angle.

```
robot_control call-tool ControlSixArmJoint --arm-joint-1 <value> --arm-joint-2  
<value> --arm-joint-3 <value> --arm-joint-4 <value> --arm-joint-5 <value> --arm-  
joint-6 <value> --runtime <value>
```

Parameter	Type	Required	Description
<code>--arm-joint-1</code>	integer	Yes	Angle for Joint 1
<code>--arm-joint-2</code>	integer	Yes	Angle for Joint 2
<code>--arm-joint-3</code>	integer	Yes	Angle for Joint 3
<code>--arm-joint-4</code>	integer	Yes	Angle for Joint 4
<code>--arm-joint-5</code>	integer	Yes	Angle for Joint 5
<code>--arm-joint-6</code>	integer	Yes	Gripper angle; range [0, 180], where 180 = fully open, 0 = fully closed.
<code>--runtime</code>	integer	No	Duration of the movement (in milliseconds); a lower value results in faster movement. Default: 1500; Range: [0, 2000].

ControlSingleArmJoint

Sets a single joint of the robotic arm to a specified angle.

```
robot_control call-tool ControlSingleArmJoint --arm-joint-id <value> --angle  
<value> --runtime <value>
```

Parameter	Type	Required	Description
<code>--arm-joint-id</code>	integer	Yes	Joint ID (1-6), corresponding to robotic arm joints 1 through 6; Joint 6 corresponds to the gripper.
<code>--angle</code>	integer	Yes	Target angle; for Joint 6 (grripper), the range is [0, 180], where 180 = fully open, 0 = fully closed.
<code>--runtime</code>	integer	No	Movement duration (milliseconds); a smaller value results in faster movement. Default: 1500. Range: [0, 2000].

Place

Place an object at the specified coordinates.

```
robot_control call-tool Place --place-x <value> --place-y <value> --place-z <value>
```

Parameter	Type	Required	Description
--place-x	number	Yes	X-coordinate (meters)
--place-y	number	Yes	Y-coordinate (meters)
--place-z	number	Yes	Z-coordinate (meters)

Pick

Pick up an object.

```
robot_control call-tool Pick --x1 <value> --y1 <value> --x2 <value> --y2 <value>
```

Parameter	Type	Required	Description
--x1	integer	Yes	X1-coordinate of the target object's top-left corner
--y1	integer	Yes	Y1-coordinate of the target object's top-left corner
--x2	integer	Yes	X2-coordinate of the target object's bottom-right corner
--y2	integer	Yes	Y2-coordinate of the target object's bottom-right corner

AdjustCameraView

Adjusts the camera's viewpoint using pitch and yaw angles, and returns the adjusted image.

```
robot_control call-tool AdjustCameraView --pitch <value> --yaw <value>
```

Parameter	Type	Required	Description
--pitch	integer	No	Pitch angle adjustment: positive = up, negative = down. Recommended step size: 5–10 degrees.
--yaw	integer	No	Yaw angle adjustment: positive = right, negative = left. Recommended step size: 5–10 degrees.

SeeWhat

Captures an image from the camera and returns the file path where the image is saved.

```
robot_control call-tool SeeWhat
```

InitArmPose

Restores the robotic arm to its preset initial pose.

```
robot_control call-tool InitArmPose
```

GetBbox

Detects a target object based on a description and returns its bounding box coordinates.

```
robot_control call-tool GetBbox --query <value>
```

Parameter	Type	Required	Description
<code>--query</code>	string	Yes	A description of the target object's position and visual features within the current view.

GetPlacePoint

Identifies the coordinates for a placement point and verifies whether it lies within the robotic arm's reachable workspace.

```
robot_control call-tool GetPlacePoint --query <value>
```

Parameter	Type	Required	Description
<code>--query</code>	string	Yes	A description of the target placement point's position and visual features.

AdjustChassisFitArmRange

Quickly adjust the mobile chassis to position the target point within the robotic arm's workspace.

```
robot_control call-tool AdjustChassisFitArmRange --x <value> --y <value> --z <value>
```

Parameter	Type	Required	Description
<code>--x</code>	number	Yes	Target X-coordinate (m)
<code>--y</code>	number	Yes	Target Y-coordinate (m)
<code>--z</code>	number	Yes	Target Z-coordinate (m)

RecordMapLocation

Save the current location to the map file.

```
robot_control call-tool RecordMapLocation --name <value> --symbol <value>
```

Parameter	Type	Required	Description
<code>--name</code>	string	Yes	Location name
<code>--symbol</code>	string	Yes	Location identifier

GetMapMapping

Retrieve the mapping between all location names and their corresponding identifiers

```
robot_control call-tool GetMapMapping
```

Navigation

Navigate to a target location using its identifier

```
robot_control call-tool Navigation --location <value>
```

Parameter	Type	Required	Description
<code>--location</code>	string	yes	The identifier corresponding to the location in the map mapping file

GetWasteRecognitionResults

Retrieve waste recognition results from the YOLO model

```
robot_control call-tool GetWasteRecognitionResults
```

GraspWaste

Grasp a waste object

```
robot_control call-tool Graspwaste --waste-name <value>
```

Parameter	Type	Required	Description
<code>--waste-name</code>	string	yes	The name of the waste object to be grasped (obtained from the waste recognition results)

PlaceWaste

Place a waste object

```
robot_control call-tool PlaceWaste
```

TTS

Robot voice broadcast (Text-to-Speech)

```
robot_control call-tool TTS --text <value>
```

Parameter	Type	Required	Description
<code>--text</code>	string	yes	The text to be broadcast

Rotate

Rotate in place by a specified angle

```
robot_control call-tool Rotate --angle <value>
```

Parameter	Type	Required	Description
<code>--angle</code>	number	yes	Rotation angle (in degrees); negative = clockwise/right, positive = counter-clockwise/left

TargetTrack

Visual tracking of a target object

```
robot_control call-tool TargetTrack --x1-or-cmd <value> --y1 <value> --x2 <value> --y2 <value>
```

Parameter	Type	Required	Description
<code>--x1-or-cmd</code>	string	yes	Enter the string "cancel" to stop tracking; enter an integer as the x1 coordinate of the target tracking bounding box (JSON string)
<code>--y1</code>	string	no	JSON string
<code>--x2</code>	string	no	JSON string
<code>--y2</code>	string	no	JSON string

GetTargetDist

Calculates the distance between a target and the robot's chassis based on the object's description.

```
robot_control call-tool GetTargetDist --query <value>
```

Parameter	Type	Required	Description
<code>--query</code>	string	yes	A description of the target object's position and visual features within the current field of view

GraspArUcoTarget

Grasps an AprilTag target with the specified ID.

```
robot_control call-tool GraspArUcoTarget --target-id <value>
```

Parameter	Type	Required	Description
<code>--target-id</code>	integer	yes	The ID of the AprilTag to be grasped

GetAprilTagIDs

Retrieves the IDs of AprilTags currently within the field of view.

```
robot_control call-tool GetAprilTagIDs
```

FollowLine

Follows a colored line on the ground.

```
robot_control call-tool FollowLine --color <value>
```

Parameter	Type	Required	Description
<code>--color</code>	string	yes	The color of the line to follow; options: red, green, blue, yellow

Useful Commands

```
robot_control list-tools
```

Command Summary Table

Command	Parameters	Function
MoveWithSpeed	<code>--linear-x</code> , <code>--linear-y</code> , <code>--angular-z</code> , <code>--duration</code>	Controls the robot's omnidirectional chassis movement via velocity commands
Move	<code>--distance</code> , <code>--direction</code>	Controls the robot's chassis to move a specified distance
GetCurrentArmJointAngles	None	Retrieves the current angles of each joint of the robotic arm
GetEndEffectorPose	None	Retrieves the pose (position and orientation) of the end effector (including the gripper and camera)
ControlSixArmJoint	<code>--arm-joint-1</code> ~ <code>--arm-joint-6</code> , <code>--runtime</code>	Simultaneously controls the 6 joints of the robotic arm; Joint 6 controls the gripper angle
ControlSingleArmJoint	<code>--arm-joint-id</code> , <code>--angle</code> , <code>--runtime</code>	Sets a single joint of the robotic arm to a specified angle
Place	<code>--place-x</code> , <code>--place-y</code> , <code>--place-z</code>	Places an object at a specified coordinate position
Pick	<code>--x1</code> , <code>--y1</code> , <code>--x2</code> , <code>--y2</code>	Picks up an object
AdjustCameraView	<code>--pitch</code> , <code>--yaw</code>	Adjusts the camera's field of view using pitch and yaw angles
SeeWhat	None	Captures an image from the camera and returns the path where the image is saved
InitArmPose	None	Restores the robotic arm to its preset initial pose
GetBbox	<code>--query</code>	Detects and returns the bounding box coordinates of a target object based on a description
GetPlacePoint	<code>--query</code>	Finds the coordinates for a placement point and checks if it lies within the robotic arm's reachable workspace

Command	Parameters	Function
AdjustChassisFitArmRange	<code>--x</code> , <code>--y</code> , <code>--z</code>	Quickly adjusts the mobile chassis to ensure the target point falls within the robotic arm's working range
RecordMapLocation	<code>--name</code> , <code>--symbol</code>	Saves the current location to the map mapping file
GetMapMapping	None	Retrieves the mapping relationship between all location names and their corresponding identifiers
Navigation	<code>--location</code>	Navigates to a target location using the target point's identifier
	GetWasteRecognitionResults	None
GraspWaste	<code>--waste-name</code>	Grasps a waste object
PlaceWaste	None	Places a waste object
TTS	<code>--text</code>	Robot voice broadcast
Rotate	<code>--angle</code>	Rotates in place by a specified angle
TargetTrack	<code>--x1-or-cmd</code> , <code>--y1</code> , <code>--x2</code> , <code>--y2</code>	Visual tracking of a target object
GetTargetDist	<code>--query</code>	Calculates the distance between a target object and the robot chassis based on an object description
GraspArUcoTarget	<code>--target-id</code>	Grasps an AprilTag target with a specified ID
GetAprilTagIDs	None	Retrieves the IDs of AprilTags currently within the field of view
FollowLine	<code>--color</code>	Follows a colored line on the ground